

1. What is Azure Databricks?

Azure Databricks is a cloud-based big data analytics platform provided by Microsoft and built on the open-source Apache Spark framework. It enables data engineers, data scientists, and analysts to process large amounts of data, build machine learning models, and perform analytics collaboratively.

Key Features:

- Fast big data processing using Spark
- Collaborative notebooks (Python, SQL, Scala, R)
- Machine Learning and AI support
- Integration with Azure services
- Scalable cloud infrastructure

2. Advantages of Using Azure Databricks

Advantage	Description
High Performance	Uses Apache Spark for fast distributed processing.
Scalability	Automatically scales resources based on workload.
Collaboration	Multiple users can work together using shared notebooks.
Azure Integration	Easily connects with Azure Data Lake, Azure SQL Database, Power BI, etc.
Machine Learning Support	Provides built-in ML libraries and frameworks.
Security	Offers enterprise-grade security and access control.
Cost Efficiency	Pay only for resources used.
Real-Time Analytics	Supports streaming and batch data processing.

3. What is a Workspace?

A **Workspace** in Azure Databricks is a collaborative environment where users can create, manage, and organize:

- Notebooks
- Libraries
- Dashboards
- Machine Learning models
- Data analysis projects

It acts as a central location where teams can work together on data engineering, analytics, and AI projects.

Functions of a Workspace:

- Store notebooks and code
- Manage users and permissions
- Share projects among team members
- Organize data analytics workflows

4. What is EDA (Exploratory Data Analysis)?

Exploratory Data Analysis (EDA) is the process of examining, summarizing, and visualizing data to understand its characteristics before applying statistical models or machine learning algorithms.

Objectives of EDA:

- Understand data structure
- Identify missing values
- Detect outliers
- Find patterns and relationships
- Check data quality
- Generate insights for decision-making

Common EDA Techniques:

- Summary statistics (mean, median, mode)
- Histograms
- Box plots
- Scatter plots
- Correlation analysis
- Data visualization

Example:

For a customer dataset, EDA can help determine:

- Average customer age
- Most popular product
- Missing customer records
- Relationship between income and spending

In Short: EDA is the first step in data analysis used to understand and prepare data for further processing and modeling.